

ForEveryOne

File

Name : Shaft3

Changed by: Matin on: 26.01.2023 at: 01:47:10

Analysis of shafts, axle and beams

Input data

Coordinate system shaft: see picture W-002

Label	Shaft 1
Drawing	
Initial position (mm)	0.000
Length (mm)	280.000
Speed (1/min)	69.70
Sense of rotation: clockwise	
Material	C45 (1)
Young's modulus (N/mm ²)	206000.000
Poisson's ratio nu	0.300
Density (kg/m ³)	7830.000
Coefficient of thermal expansion (10 ⁻⁶ /K)	11.500
Temperature (°C)	20.000
Weight of shaft (kg)	2.755
(Notice: Weight stands for the shaft only without considering the gears)	
Weight of shaft, including additional masses (kg)	20.671
Mass moment of inertia (kg*m ²)	0.019
Momentum of mass GD2 (Nm ²)	0.742
Position in space (°)	0.000
Gears mounted with stiffness according to ISO	
Consider deformations due to shearing	
Shear correction factor	1.100
Contact angle of rolling bearings is considered	
Tolerance field: Mean value	
Reference temperature (°C)	20.000

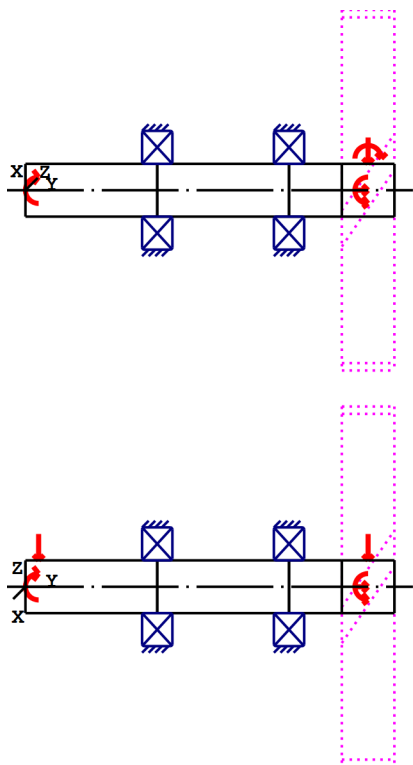


Figure: Load applications

Shaft definition (Shaft 1)

Outer contour

Cylinder (Cylinder)			0.000mm ...	100.000mm
Diameter (mm)	[d]	40.0000		
Length (mm)	[l]	100.0000		
Surface roughness (µm)	[Rz]	8.0000		
Cylinder (Cylinder)			100.000mm ...	200.000mm
Diameter (mm)	[d]	40.0000		
Length (mm)	[l]	100.0000		
Surface roughness (µm)	[Rz]	8.0000		
Cylinder (Cylinder)			200.000mm ...	280.000mm
Diameter (mm)	[d]	40.0000		
Length (mm)	[l]	80.0000		
Surface roughness (µm)	[Rz]	8.0000		

Forces

Type of force element		Cylindrical gear
Label in the model		Cylindrical gear
Position on shaft (mm)	[Ylocal]	260.0000
Position in global system (mm)	[Yglobal]	260.0000
Operating pitch diameter (mm)		273.2338
Helix angle (°)		25.1456 left
Working pressure angle at normal section (°)		20.8359

Position of contact (°)	0.0000
Length of load application (mm)	39.8780
Power (kW)	3.0000 driven (input)
Torque (Nm)	411.0171
Axial force (N)	1412.2221
Shearing force X (N)	-1264.8662
Shearing force Z (N)	-3008.5372
Bending moment X (Nm)	-0.0000
Bending moment Z (Nm)	192.9334

Type of force element		Rope sheave
Label in the model		Rope sheave
Position on shaft (mm)	[ylocal]	10.0000
Position in global system (mm)	[yglobal]	10.0000
Direction of belt force (°)		-90.0000
Sheave diameter (mm)		0.0000
Length of load application (mm)		20.0000
Power (kW)		3.0000 driving (output)
Torque (Nm)		-411.0171
Axial force (N)		0.0000
Shearing force X (N)		0.0000
Shearing force Z (N)		-2500.0000
Bending moment X (Nm)		0.0000
Bending moment Z (Nm)		0.0000
Sum of the belt forces (N)		2500.0000

Bearing

Label in the model		Rolling bearing
Bearing type		SKF 6308
Bearing type		Deep groove ball bearing (single row)
		SKF EXPLORER
Bearing position (mm)	[ylocal]	200.000
Bearing position (mm)	[yglobal]	200.000
Attachment of external ring		Set fixed bearing right
Inner diameter (mm)	[d]	40.000
External diameter (mm)	[D]	90.000
Width (mm)	[b]	23.000
Corner radius (mm)	[r]	1.500
Basic static load rating (kN)	[C ₀]	24.000
Basic dynamic load rating (kN)	[C]	42.300
Fatigue load rating (kN)	[C _u]	1.000
Values for approximated geometry:		
Basic dynamic load rating (kN)	[C _{theo}]	0.000
Basic static load rating (kN)	[C _{0theo}]	0.000
Correction factor Basic dynamic load rating	[f _C]	1.000
Correction factor Basic static load rating	[f _{C0}]	1.000

Label in the model		Rolling bearing
Bearing type		SKF 6308
Bearing type		Deep groove ball bearing (single row)
		SKF EXPLORER

Bearing position (mm)	[y _{lokal}]	100.000
Bearing position (mm)	[y _{global}]	100.000
Attachment of external ring		Set fixed bearing left
Inner diameter (mm)	[d]	40.000
External diameter (mm)	[D]	90.000
Width (mm)	[b]	23.000
Corner radius (mm)	[r]	1.500
Basic static load rating (kN)	[C ₀]	24.000
Basic dynamic load rating (kN)	[C]	42.300
Fatigue load rating (kN)	[C _u]	1.000
Values for approximated geometry:		
Basic dynamic load rating (kN)	[C _{theo}]	0.000
Basic static load rating (kN)	[C _{0theo}]	0.000
Correction factor Basic dynamic load rating	[f _C]	1.000
Correction factor Basic static load rating	[f _{C0}]	1.000

Shaft 'Shaft 1': Cylindrical gear 'Cylindrical gear' (y= 260.0000 (mm)) is taken into account as component of the shaft.
EI (y= 240.0610 (mm)): 25886.7235 (Nm²), EI (y= 279.9390 (mm)): 25886.7235 (Nm²), m (yS= 260.0000 (mm)): 17.9162 (kg)
Jp: 0.0184 (kg*m²), Jxx: 0.0099 (kg*m²), Jzz: 0.0099 (kg*m²)

Results

Shaft

Maximum deflection (µm)	81.006
Position of the maximum (mm)	280.000
Mass center of gravity (mm)	140.000
Total axial load (N)	1412.222
Torsion under torque (°)	0.272

Bearing

Probability of failure	[n]	10.00	%
Axial clearance	[u _A]	10.00	µm
Lubricant	Oil: ISO-VG 220		
Lubricant - service temperature	[T _B]	20.00	°C
Rolling bearings, classical calculation (contact angle considered)			

Shaft 'Shaft 1' Rolling bearing 'Rolling bearing'

Position (Y-coordinate)	[y]	200.00	mm
Dynamic equivalent load	[P]	4.77	kN
Equivalent load	[P ₀]	4.77	kN
Life modification factor for reliability[a ₁]		1.000	
Basic bearing rating life	[L _{nh}]	167174.17	h
Operating viscosity	[v]	912.87	mm²/s
Static safety factor	[S ₀]	5.04	

Bearing reaction force	[Fx]	3.953	kN
Bearing reaction force	[Fy]	-1.412	kN
Bearing reaction force	[Fz]	2.662	kN
Bearing reaction force	[Fr]	4.766	kN (33.96°)
Bearing reaction moment	[Mx]	0.00	Nm
Bearing reaction moment	[My]	0.00	Nm
Bearing reaction moment	[Mz]	-0.00	Nm
Bearing reaction moment	[Mr]	0.00	Nm (-56.31°)
Oil level	[H]	0.000	mm
Rolling moment of friction	[M _{rr}]	0.178	Nm
Sliding moment of friction	[M _{sl}]	0.116	Nm
Moment of friction, seals	[M _{seal}]	0.000	Nm
Moment of friction for seals determined according to SKF main catalog 17000/1 EN:2018			
Moment of friction flow losses	[M _{drag}]	0.000	Nm
Torque of friction	[M _{loss}]	0.294	Nm
Power loss	[P _{loss}]	2.146	W

The moment of friction is calculated according to the details in SKF Catalog 2018.

The calculation is always performed with a coefficient for additives in the lubricant $\mu_{bl}=0.15$.

Displacement of bearing	[u _x]	-5.468	μm
Displacement of bearing	[u _y]	10.000	μm
Displacement of bearing	[u _z]	-3.515	μm
Displacement of bearing	[u _r]	6.500	μm (-147.26°)
Misalignment of bearing	[r _x]	-0.363	mrاد (-1.25')
Misalignment of bearing	[r _y]	3.922	mrاد (13.48')
Misalignment of bearing	[r _z]	0.472	mrاد (1.62')
Misalignment of bearing	[r _r]	0.596	mrاد (2.05')

Shaft 'Shaft 1' Rolling bearing 'Rolling bearing'

Position (Y-coordinate)	[y]	100.00	mm
Dynamic equivalent load	[P]	3.98	kN
Equivalent load	[P ₀]	3.98	kN
Life modification factor for reliability[a ₁]		1.000	
Basic bearing rating life	[L _{nh}]	288143.00	h
Operating viscosity	[v]	912.87	mm ² /s
Static safety factor	[S ₀]	6.04	
Bearing reaction force	[Fx]	-2.688	kN
Bearing reaction force	[Fy]	0.000	kN
Bearing reaction force	[Fz]	2.928	kN
Bearing reaction force	[Fr]	3.975	kN (132.55°)
Bearing reaction moment	[Mx]	0.00	Nm
Bearing reaction moment	[My]	0.00	Nm
Bearing reaction moment	[Mz]	-0.00	Nm
Bearing reaction moment	[Mr]	0.00	Nm (-50.19°)
Oil level	[H]	0.000	mm
Rolling moment of friction	[M _{rr}]	0.084	Nm
Sliding moment of friction	[M _{sl}]	0.048	Nm
Moment of friction, seals	[M _{seal}]	0.000	Nm
Moment of friction for seals determined according to SKF main catalog 17000/1 EN:2018			
Moment of friction flow losses	[M _{drag}]	0.000	Nm
Torque of friction	[M _{loss}]	0.132	Nm
Power loss	[P _{loss}]	0.964	W

The moment of friction is calculated according to the details in SKF Catalog 2018.

The calculation is always performed with a coefficient for additives in the lubricant $\mu_{bl}=0.15$.

Displacement of bearing	[u _x]	4.170	μm
Displacement of bearing	[u _y]	10.000	μm
Displacement of bearing	[u _z]	-4.986	μm

Displacement of bearing	$[u_r]$	6.500	μm (-50.1°)
Misalignment of bearing	$[r_x]$	0.428	mrad (1.47')
Misalignment of bearing	$[r_y]$	1.858	mrad (6.39')
Misalignment of bearing	$[r_z]$	-0.047	mrad (-0.16')
Misalignment of bearing	$[r_r]$	0.431	mrad (1.48')

Damage (%)			[Lreq] (20000.000)
Bin no	B1	B2	
1	11.96	6.94	

Σ	11.96	6.94	

Utilization (%)			[Lreq] (20000.000)
B1	B2		
49.27	41.10		
Note: Utilization = $(L_{req}/L_h)^{(1/k)}$			
Ball bearing: $k = 3$, roller bearing: $k = 10/3$			

B1: Rolling bearing
B2: Rolling bearing

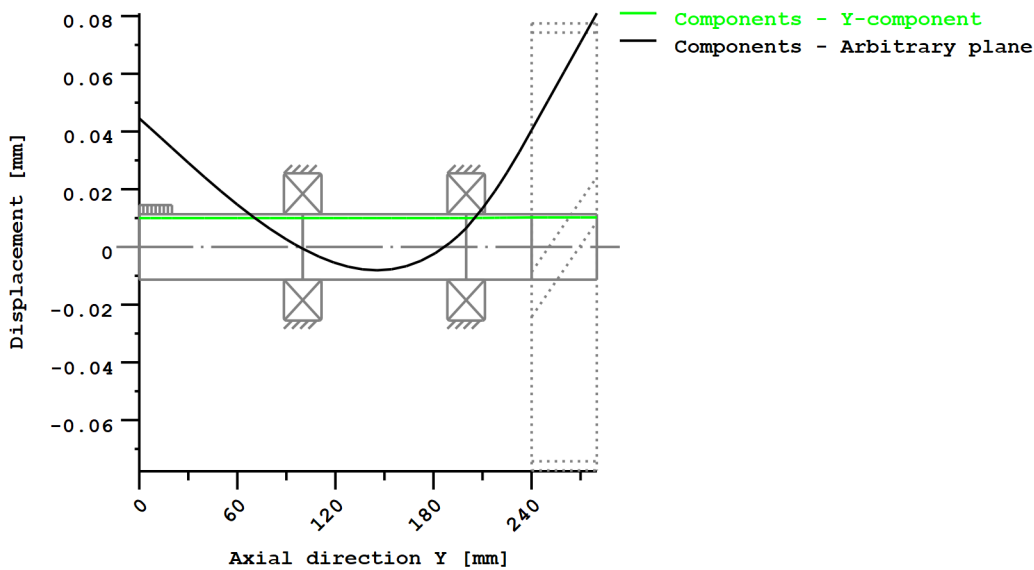
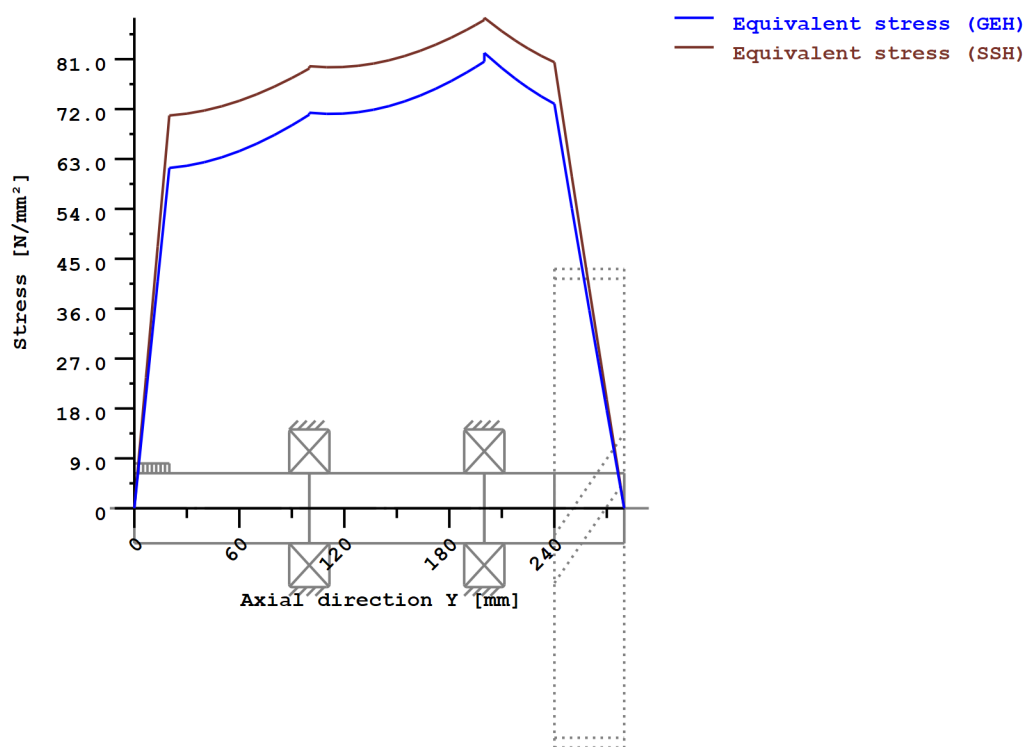


Figure: Deformation (bending etc.) (Arbitrary plane 214.5716911 121)



Nominal stresses, without taking into account stress concentrations
GEH(von Mises): $\text{sigV} = ((\text{sigB} + \text{sigZ}, D)^2 + 3 * (\text{tauT} + \text{tauS})^2)^{1/2}$
SSH(Tresca): $\text{sigV} = ((\text{sigB} - \text{sigZ}, D)^2 + 4 * (\text{tauT} + \text{tauS})^2)^{1/2}$

Figure: Equivalent stress